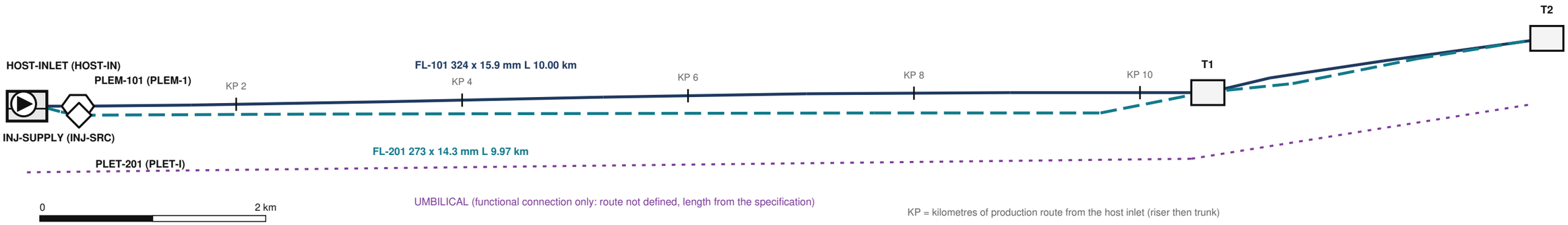
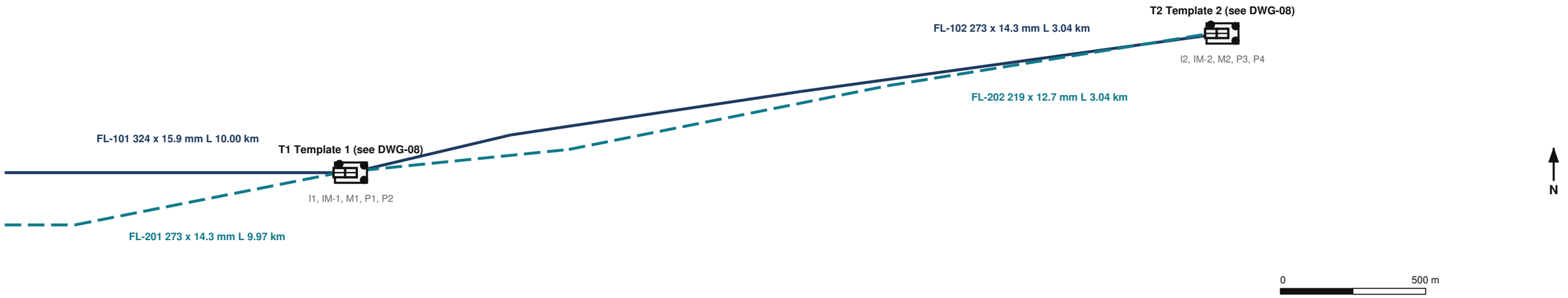
SURF-SYN-001-DWG-01 Overall field layout plan

Reference concept A (Concept A - daisy chain). Coordinates: local synthetic grid (m); vertical datum mean sea level.

OVERALL PLAN 1:36 600



TEMPLATE AREA DETAIL 1:14 300



SCREENING - screening grade: synthetic inputs, 8 open specialist hand-off(s), 5 data gap(s)

LEGEND

- production flowline, jumper, riser
- injection flowline, jumper, riser
- umbilical (functional)
- template footprint (to scale in the detail)
- PLEM / PLET / host symbols (exaggerated)

GENERAL NOTES

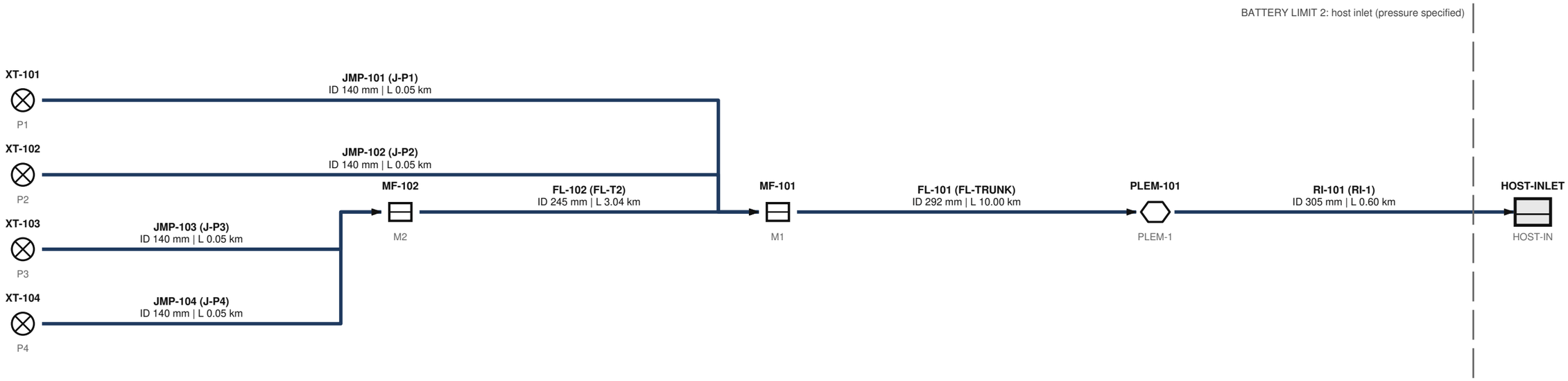
1. Plan view of the reference concept from the canonical dataset; line coordinates are the specification route points (synthetic, not surveyed).
2. The umbilical is drawn as a functional connection between the host and the template terminations; its route and cross-section are not defined.
3. Symbols are exaggerated for legibility; positions are accurate only for line geometry and template footprints at the stated scales.
4. Issue purpose: Issued for review (diagnostic, synthetic example); readiness SCREENING (assigned by the toolkit gates, not by a person).
5. Run reference: dataset 7a2b4b0b67d3d98726e0d58d, NeqSim workspace classes, study SURF-SYN.
6. Units: pressure bara, temperature degC, lengths m, rates kg/s. Source/model version: surf-study 1.0.

Rev	Date	Description
01	2026-10-25	First generated issue from surf_study.v1

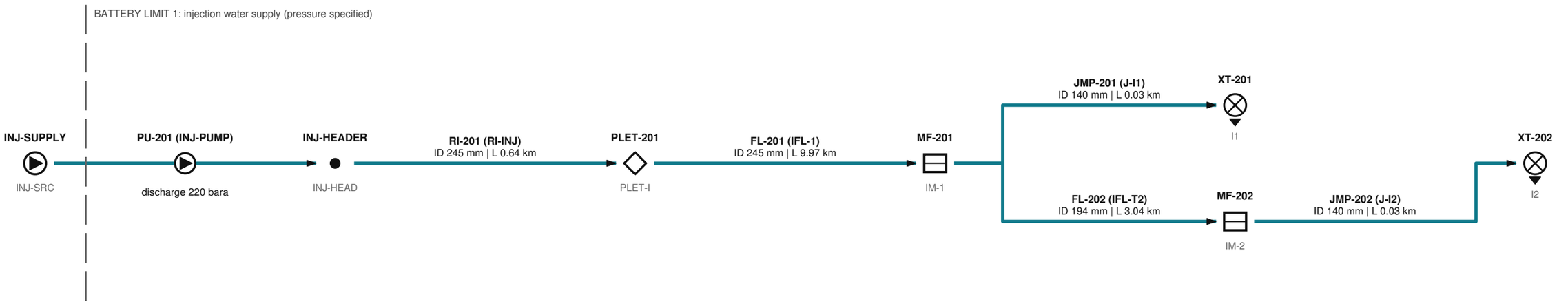
Company: NeqSim open example		Project: Synthetic SURF demonstration	
Title: **Four-well subsea tie-back with water injection: SURF design study (synthetic example) - Overall field layout plan**			
Drawing number: **SURF-SYN-001-DWG-01**		Revision: 01	Status: **SCREENING**
Sheet: Sheet 1 of 15	Date: 2026-10-25	Scale: overall 1:36 600, detail 1:14 300 (equal x ...	Format: A3 landscape
Drawn by: surf-study toolkit	Checked by: not assigned	Approved by: not assigned	

SURF-SYN-001-DWG-02 SURF process and network schematic

PRODUCTION NETWORK (concept A)



INJECTION NETWORK



SCREENING - screening grade: synthetic inputs, 8 open specialist hand-off(s), 5 data gap(s)

LEGEND

- production line (arrow = modelled flow direction)
- injection line
- well / manifold / PLEM / PLET / host symbols

GENERAL NOTES

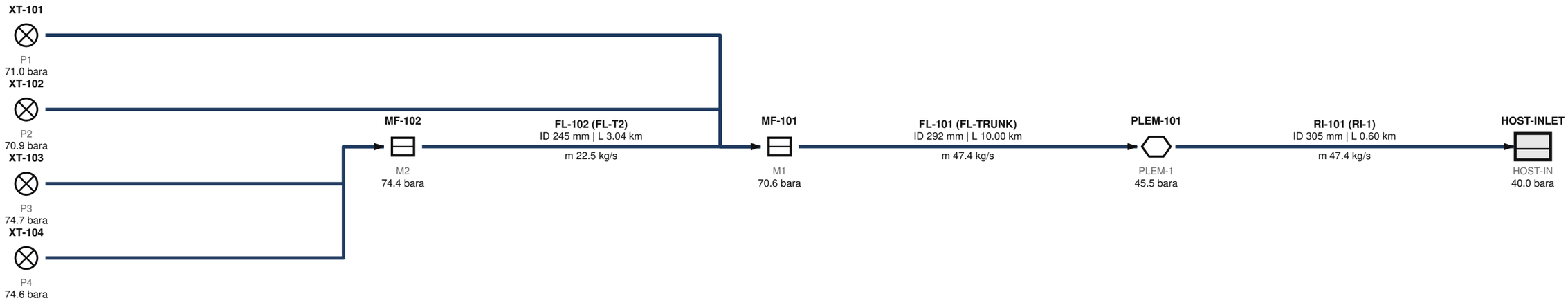
1. Connectivity is read from the generated NeqSim topology (FieldNetworkTopology over LoopedPipeNetwork); each tag and size is the canonical dataset value.
2. Production chokes, isolation valves and instrumentation are not modelled and are not shown; see the functional sheets for omitted systems.
3. The two services are separate hydraulic domains of the same canonical identity registry (different fluids).
4. Issue purpose: Issued for review (diagnostic, synthetic example); readiness SCREENING (assigned by the toolkit gates, not by a person).
5. Run reference: dataset 7a2b4b0b67d3d98726e0d58d, NeqSim workspace classes, study SURF-SYN.
6. Units: pressure bara, temperature degC, lengths m, rates kg/s. Source/model version: surf-study 1.0.

Rev	Date	Description
01	2026-10-25	First generated issue from surf_study.v1

Company: NeqSim open example		Project: Synthetic SURF demonstration	
Title: **Four-well subsea tie-back with water injection: SURF design study (synthetic example) - SURF process and network schematic**			
Drawing number: **SURF-SYN-001-DWG-02**		Revision: 01	Status: **SCREENING**
Sheet: Sheet 2 of 15	Date: 2026-10-25	Scale: see sheet notes	Format: A3 landscape
Drawn by: surf-study toolkit	Checked by: not assigned	Approved by: not assigned	

SURF-SYN-001-DWG-03 Production process-flow and functional sheet

PRODUCTION - design case BASE (Plateau, base case)



Operating conditions at the modelled nodes: p in bara, T in degC (solved values, all cases)

Node	Tag	BASE p	BASE T	TURNDOWN p	TURNDOWN T	LATE p	LATE T
HOST-IN	HOST-INLET	40.0	15.0	60.0	15.0	35.0	15.0
M1	MF-101	70.6	15.0	77.5	15.0	55.3	15.0
M2	MF-102	74.4	15.0	78.6	15.0	56.0	15.0
P1	XT-101	71.0	80.0	77.6	80.0	55.3	80.0
P2	XT-102	70.9	80.0	77.6	80.0	55.3	80.0
P3	XT-103	74.7	80.0	78.7	80.0	56.0	80.0
P4	XT-104	74.6	80.0	78.7	80.0	56.0	80.0
PLEM-1	PLEM-101	45.5	15.0	72.1	15.0	52.0	15.0

OMITTED SYSTEMS (functional schematic, not an issued P&ID)

- Production/injection chokes and tree valves (master, wing, swab): not modelled
- Isolation valves, ESD and depressurisation valves: not modelled
- Pressure, temperature and flow instrumentation and the control system: not modelled
- Chemical injection points and the umbilical service tubes: see DWG-10
- Safety functions, SIL, valve closure times and cause-and-effect: not defined

SCREENING - screening grade: synthetic inputs, 8 open specialist hand-off(s), 5 data gap(s)

LEGEND

- modelled line with flow direction
- modelled node symbols; values are the BASE case

GENERAL NOTES

1. Functional process-flow schematic: only items present in the generated model are drawn; omitted systems are listed on the sheet.
2. This is not an issued-for-construction P&ID: instrumentation, valves and safety design are absent from the study scope.
3. Issue purpose: Issued for review (diagnostic, synthetic example); readiness SCREENING (assigned by the toolkit gates, not by a person).
4. Run reference: dataset 7a2b4b0b67d3d98726e0d58d, NeqSim workspace classes, study SURF-SYN.
5. Units: pressure bara, temperature degC, lengths m, rates kg/s. Source/model version: surf-study 1.0.

Rev	Date	Description
01	2026-10-25	First generated issue from surf_study.v1

Company: NeqSim open example		Project: Synthetic SURF demonstration	
Title: **Four-well subsea tie-back with water injection: SURF design study (synthetic example) - Production process-flow and functional sheet**			
Drawing number: **SURF-SYN-001-DWG-03**		Revision: 01	Status: **SCREENING**
Sheet: Sheet 3 of 15	Date: 2026-10-25	Scale: see sheet notes	Format: A3 landscape
Drawn by: surf-study toolkit	Checked by: not assigned	Approved by: not assigned	

SURF-SYN-001-DWG-04 Injection process-flow and functional sheet

INJECTION - design case BASE (Plateau, base case)



Operating conditions at the modelled nodes: p in bara, T in degC (solved values, all cases)

Node	Tag	BASE p	BASE T	TURNDOWN p	TURNDOWN T	LATE p	LATE T
I1	XT-201	250.5	30.0	233.8	30.0	263.8	30.0
I2	XT-202	250.6	30.0	234.4	30.0	264.4	30.0
IM-1	MF-201	250.7	30.0	233.9	30.0	263.9	30.0
IM-2	MF-202	250.7	30.0	234.5	30.0	264.5	30.0
INJ-SRC	INJ-SUPPLY	10.0	30.0	10.0	30.0	10.0	30.0
PLET-I	PLET-201	255.9	30.0	236.1	30.0	266.1	30.0

OMITTED SYSTEMS (functional schematic, not an issued P&ID)

- Production/injection chokes and tree valves (master, wing, swab): not modelled
- Isolation valves, ESD and depressurisation valves: not modelled
- Pressure, temperature and flow instrumentation and the control system: not modelled
- Chemical injection points and the umbilical service tubes: see DWG-10
- Safety functions, SIL, valve closure times and cause-and-effect: not defined

SCREENING - screening grade: synthetic inputs, 8 open specialist hand-off(s), 5 data gap(s)

LEGEND

- modelled line with flow direction
- modelled node symbols; values are the BASE case

GENERAL NOTES

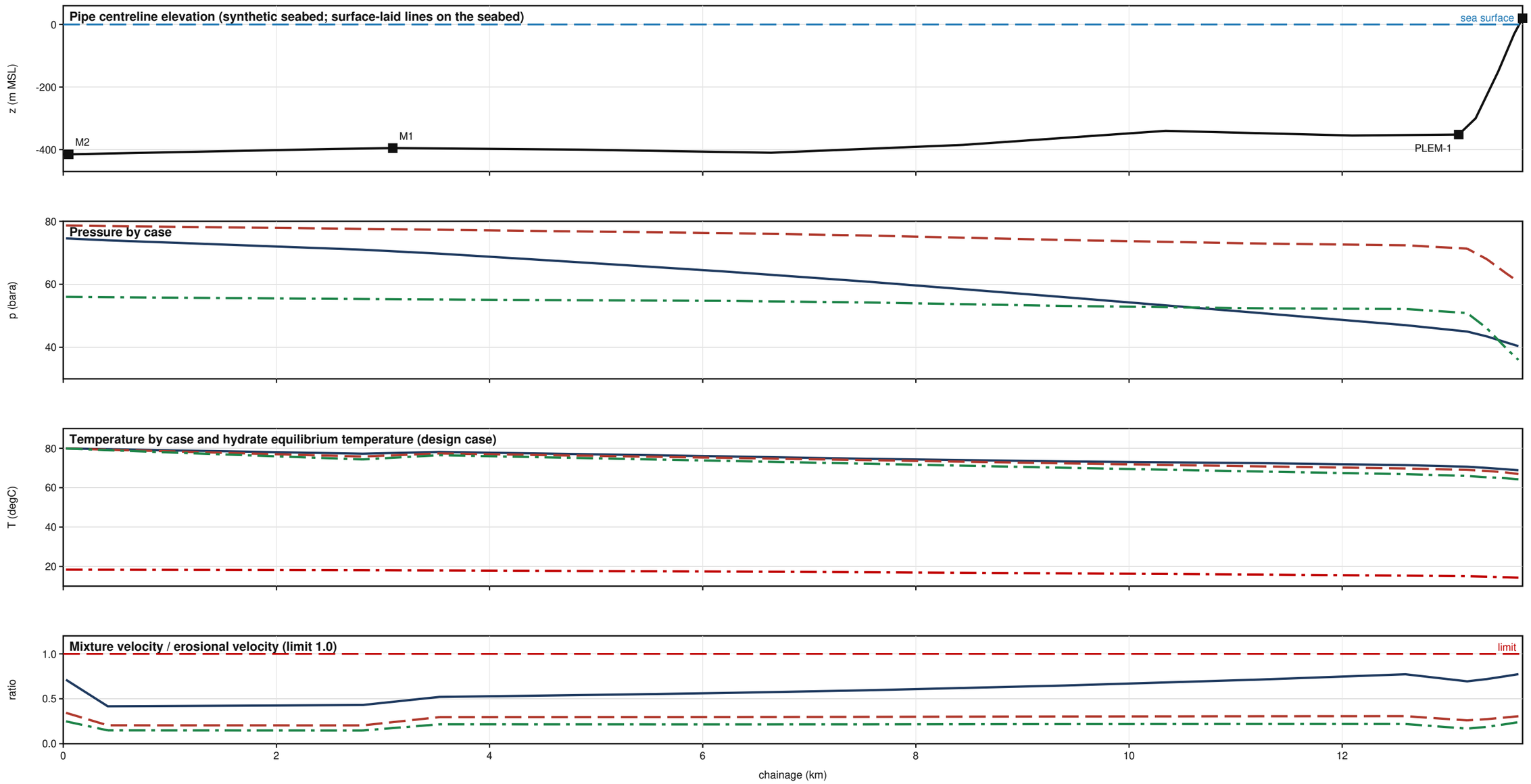
1. Functional process-flow schematic: only items present in the generated model are drawn; omitted systems are listed on the sheet.
2. This is not an issued-for-construction P&ID: instrumentation, valves and safety design are absent from the study scope.
3. Issue purpose: Issued for review (diagnostic, synthetic example); readiness SCREENING (assigned by the toolkit gates, not by a person).
4. Run reference: dataset 7a2b4b0b67d3d98726e0d58d, NeqSim workspace classes, study SURF-SYN.
5. Units: pressure bara, temperature degC, lengths m, rates kg/s. Source/model version: surf-study 1.0.

Rev	Date	Description
01	2026-10-25	First generated issue from surf_study.v1

Company: NeqSim open example	Project: Synthetic SURF demonstration
Title: **Four-well subsea tie-back with water injection: SURF design study (synthetic example) - Injection process-flow and functional sheet**	
Drawing number: **SURF-SYN-001-DWG-04**	Revision: 01 / Status: **SCREENING**
Sheet: Sheet 4 of 15 / Date: 2026-10-25	Scale: see sheet notes / Format: A3 landscape
Drawn by: surf-study toolkit	Checked by: not assigned / Approved by: not assigned

SURF-SYN-001-DWG-05 Production route longitudinal profile

Path J-P3 > FL-T2 > FL-TRUNK > RI-1 (longest production path; chainage measured in the flow direction from the first node of the path)



SCREENING - screening grade: synthetic inputs, 8 open specialist hand-off(s), 5 data gap(s)

LEGEND

- BASE: Plateau, base case
- TURNDOWN: Turndown (back-pressured host, reduced productivity)
- LATE: Late life, high water cut
- hydrate equilibrium temperature (CPA, uninhibited)

GENERAL NOTES

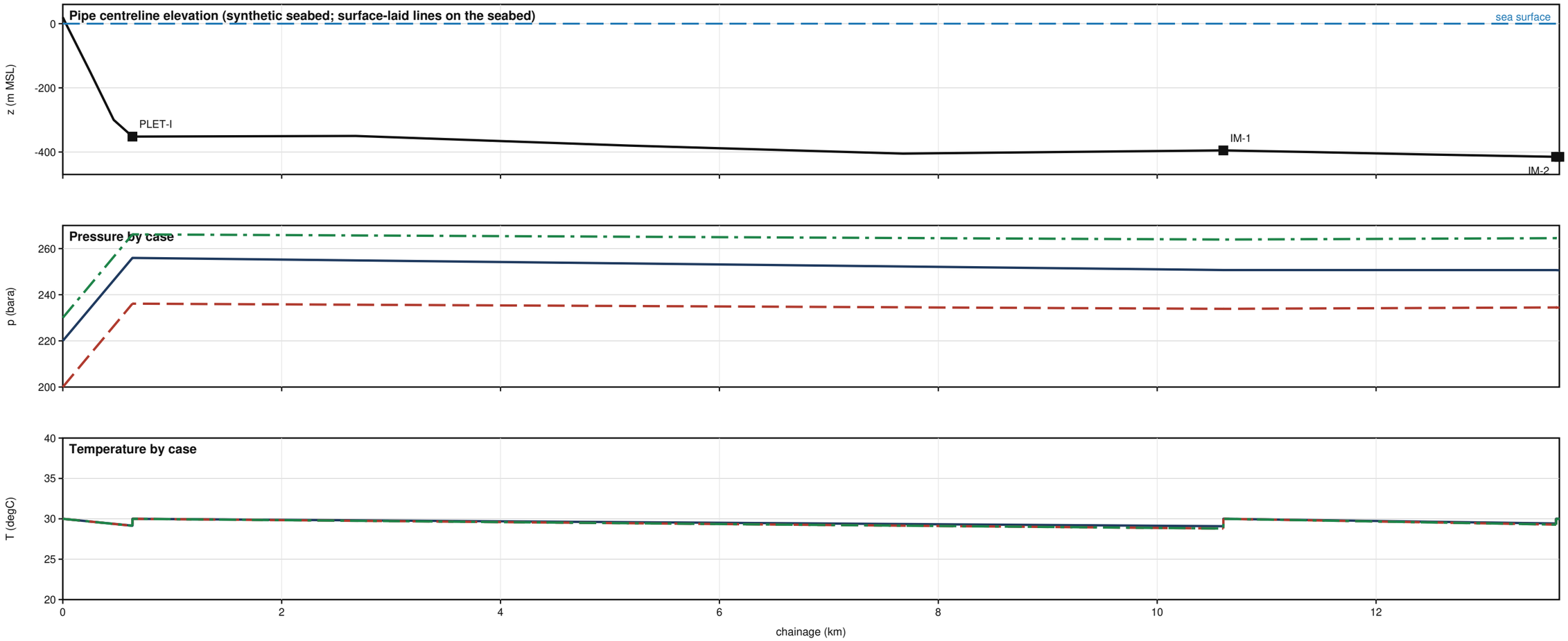
1. Profiles are extracted once from the solved cases of the reference concept; the same chainage runs through all panels.
2. Hydraulic and thermal profiles are steady-state screening results (Beggs and Brill multiphase); the seabed profile is synthetic, not surveyed.
3. Horizontal scale is the chainage axis; the vertical scale of each panel is stated by its axis and is exaggerated.
4. Issue purpose: Issued for review (diagnostic, synthetic example); readiness SCREENING (assigned by the toolkit gates, not by a person).
5. Run reference: dataset 7a2b4b0b67d3d98726e0d58d, NeqSim workspace classes, study SURF-SYN.
6. Units: pressure bara, temperature degC, lengths m, rates kg/s. Source/model version: surf-study 1.0.

Rev	Date	Description
01	2026-10-25	First generated issue from surf_study.v1

Company: NeqSim open example	Project: Synthetic SURF demonstration
Title: **Four-well subsea tie-back with water injection: SURF design study (synthetic example) - Production route longitudinal profile**	
Drawing number: **SURF-SYN-001-DWG-05**	Revision: 01 / Status: **SCREENING**
Sheet: Sheet 5 of 15 / Date: 2026-10-25	Scale: horizontal: chainage 1:41 000; vertical ex... / Format: A3 landscape
Drawn by: surf-study toolkit	Checked by: not assigned / Approved by: not assigned

SURF-SYN-001-DWG-06 Injection route longitudinal profile

Path INJ-PUMP > RI-INJ > IFL-1 > IFL-T2 > J-I2 (longest injection path; chainage measured in the flow direction from the first node of the path)



Single-phase liquid injection: profiles show the end values of each edge (the network gives the distributed profile only for multiphase edges).

SCREENING - screening grade: synthetic inputs, 8 open specialist hand-off(s), 5 data gap(s)

LEGEND

- BASE: Plateau, base case
- TURNDOWN: Turndown (back-pressured host, reduced productivity)
- LATE: Late life, high water cut

GENERAL NOTES

1. Profiles are extracted once from the solved cases of the reference concept; the same chainage runs through all panels.
2. Hydraulic and thermal profiles are steady-state screening results (Beggs and Brill multiphase); the seabed profile is synthetic, not surveyed.
3. Horizontal scale is the chainage axis; the vertical scale of each panel is stated by its axis and is exaggerated.
4. Issue purpose: Issued for review (diagnostic, synthetic example); readiness SCREENING (assigned by the toolkit gates, not by a person).
5. Run reference: dataset 7a2b4b0b67d3d98726e0d58d, NeqSim workspace classes, study SURF-SYN.
6. Units: pressure bara, temperature degC, lengths m, rates kg/s. Source/model version: surf-study 1.0.

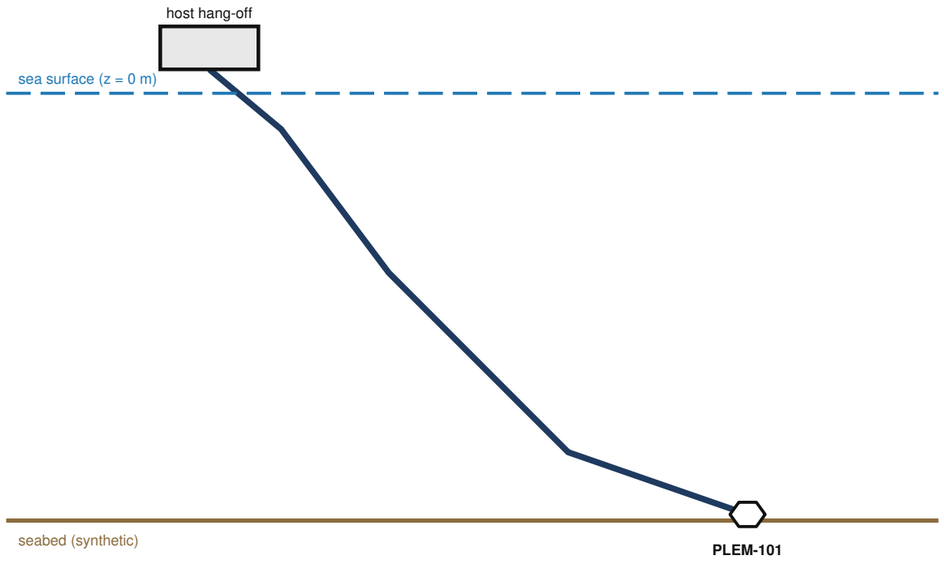
Rev	Date	Description
01	2026-10-25	First generated issue from surf_study.v1

Company: NeqSim open example	Project: Synthetic SURF demonstration
Title: **Four-well subsea tie-back with water injection: SURF design study (synthetic example) - Injection route longitudinal profile**	
Drawing number: **SURF-SYN-001-DWG-06**	Revision: 01 / Status: **SCREENING**
Sheet: Sheet 6 of 15 / Date: 2026-10-25	Scale: horizontal: chainage 1:41 000; vertical ex... / Format: A3 landscape
Drawn by: surf-study toolkit	Checked by: not assigned / Approved by: not assigned

SURF-SYN-001-DWG-07 Riser configuration and elevation (concept schematic)

RI-101 RI-1 (flexible, production)

CONCEPT SCHEMATIC - GEOMETRY NOT ANALYZED

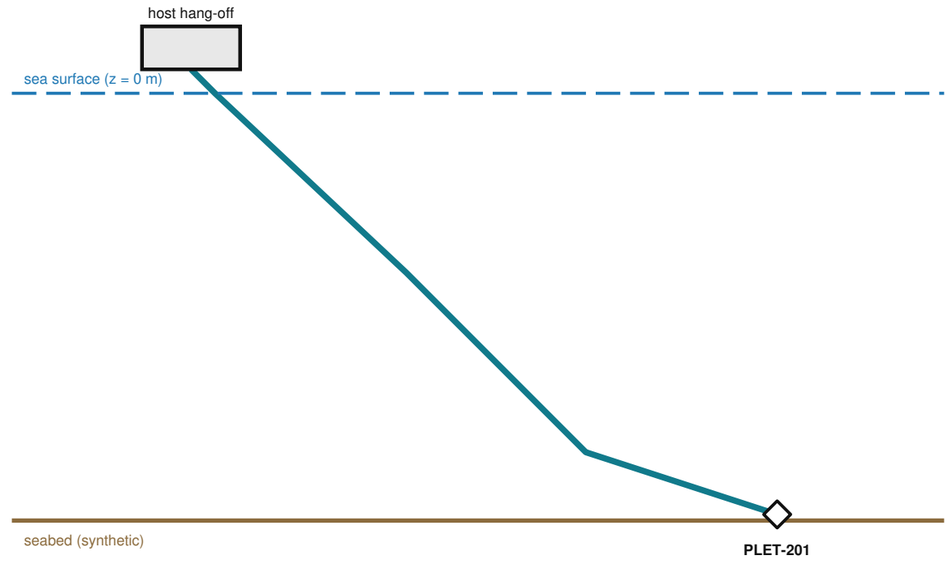


Dimensions (from the specification): installed length 599 m; horizontal offset 450 m; vertical span 372 m;
bore 305 mm; design 345 bara / 120 degC; touchdown, buoyancy and hang-off loads: not defined (analysis hand-off HO-01).

Fluid state at the riser (interface load input)	Value
Inlet pressure (bara)	45.5
Outlet pressure (bara)	40.0
Outlet temperature (degC)	68.7
Flow (kg/s, BASE case)	47.4

RI-201 RI-INJ (steel, injection)

CONCEPT SCHEMATIC - GEOMETRY NOT ANALYZED



Dimensions (from the specification): installed length 636 m; horizontal offset 500 m; vertical span 372 m;
bore 245 mm; design 345 bara / 80 degC; touchdown, buoyancy and hang-off loads: not defined (analysis hand-off HO-01).

Fluid state at the riser (interface load input)	Value
Inlet pressure (bara)	220.0
Outlet pressure (bara)	255.9
Outlet temperature (degC)	29.2
Flow (kg/s, BASE case)	81.5

SCREENING - screening grade: synthetic inputs, 8 open specialist hand-off(s), 5 data gap(s)

LEGEND

- production riser polyline
- injection riser polyline
- sea surface

GENERAL NOTES

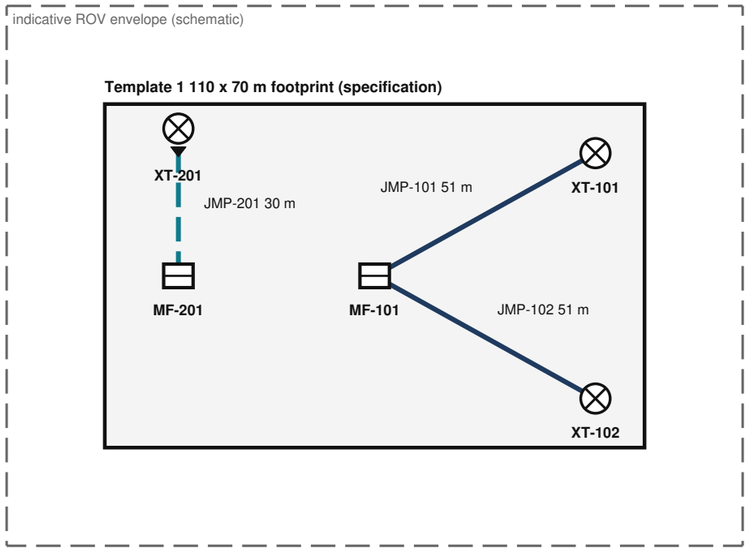
1. The riser shapes are the specification polylines. They are not an equilibrium solution: a lazy-wave or catenary label does not define the geometry.
2. Static and dynamic response, VIV and fatigue, interference and coupled host/mooring analysis require a qualified external solver (hand-off HO-01).
3. Fluid states are the solved NeqSim values and are inputs to that analysis.
4. Issue purpose: Issued for review (diagnostic, synthetic example); readiness SCREENING (assigned by the toolkit gates, not by a person).
5. Run reference: dataset 7a2b4b0b67d3d98726e0d58d, NeqSim workspace classes, study SURF-SYN.
6. Units: pressure bara, temperature degC, lengths m, rates kg/s. Source/model version: surf-study 1.0.

Rev	Date	Description
01	2026-10-25	First generated issue from surf_study.v1

Company: NeqSim open example		Project: Synthetic SURF demonstration	
Title: **Four-well subsea tie-back with water injection: SURF design study (synthetic example) - Riser configuration and elevation (concept schematic)**			
Drawing number: **SURF-SYN-001-DWG-07**		Revision: 01	Status: **SCREENING**
Sheet: Sheet 7 of 15	Date: 2026-10-25	Scale: NOT TO SCALE (concept schematic; pol...	Format: A3 landscape
Drawn by: surf-study toolkit	Checked by: not assigned	Approved by: not assigned	

SURF-SYN-001-DWG-08 Template, manifold, PLEM and PLET arrangement

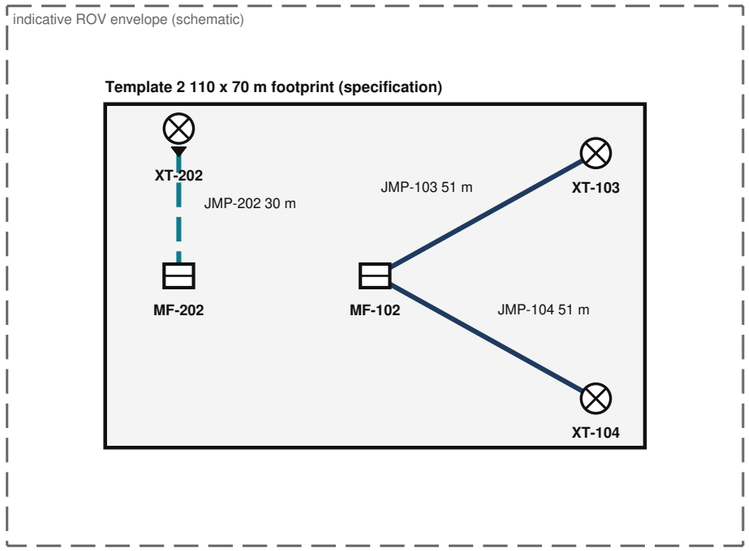
T1: plan



Plan scale about 1:1250 on the A3 sheet; schematic envelope only, not a vendor general arrangement.

Hub connections: hub 'in-daisy' < FL-102; hub 'out-trunk' > FL-101; hub 'in' < FL-201; hub 'out-T2' > FL-202

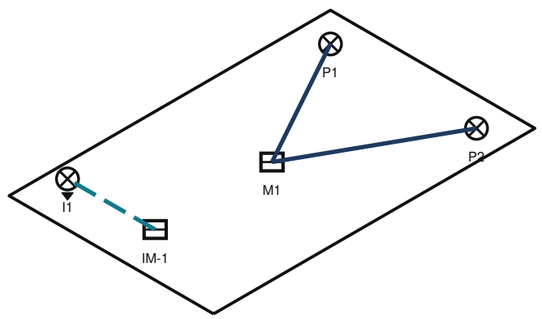
T2: plan



Plan scale about 1:1250 on the A3 sheet; schematic envelope only, not a vendor general arrangement.

Hub connections: hub 'out-daisy' > FL-102; hub 'in' < FL-202

T1: isometric (NOT TO SCALE)



All T1 equipment stands on the seabed at z = -395 m; hub orientations and heights are not defined (vendor data).

PLEM / PLET connection schematic (NOT TO SCALE)

PLEM-101 (PLEM-1)



z = -352 m; elevation head, hubs and mud-mat not defined

FL-101 in port 'in-trunk'

RI-101 out port 'out-riser'

PLET-201 (PLET-I)



z = -352 m; elevation head, hubs and mud-mat not defined

RI-201 in port 'in-riser'

FL-201 out port 'out'

SCREENING - screening grade: synthetic inputs, 8 open specialist hand-off(s), 5 data gap(s)

LEGEND

- well, manifold, PLEM, PLET symbols (exaggerated)
- production jumper
- injection jumper
- indicative ROV envelope

GENERAL NOTES

1. Positions and jumper lengths are specification values; footprints and envelopes are schematic and distinct from vendor general arrangements and fabrication drawings.
2. Hub orientation, slot geometry, maintenance and ROV envelopes are not defined; they remain vendor and installation inputs.
3. Dimensions are presented as the lengths of the registered lines (ISO 129-1 style annotation without tolerances).
4. Issue purpose: Issued for review (diagnostic, synthetic example); readiness SCREENING (assigned by the toolkit gates, not by a person).
5. Run reference: dataset 7a2b4b0b67d3d98726e0d58d, NeqSim workspace classes, study SURF-SYN.
6. Units: pressure bara, temperature degC, lengths m, rates kg/s. Source/model version: surf-study 1.0.

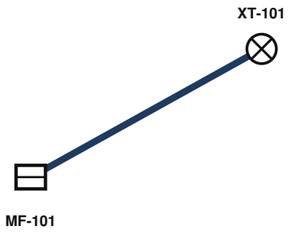
Rev	Date	Description
01	2026-10-25	First generated issue from surf_study.v1

Company: NeqSim open example		Project: Synthetic SURF demonstration	
Title: **Four-well subsea tie-back with water injection: SURF design study (synthetic example) - Template, manifold, PLEM and PLET arrangement**			
Drawing number: **SURF-SYN-001-DWG-08**		Revision: 01	Status: **SCREENING**
Sheet: Sheet 8 of 15	Date: 2026-10-25	Scale: plan panels 1:1250 (equal x and y); elevat...	Format: A3 landscape
Drawn by: surf-study toolkit	Checked by: not assigned	Approved by: not assigned	

SURF-SYN-001-DWG-09 Jumper and tie-in details

JMP-101 J-P1: XT-101 to MF-101 (production)

Plan view



horizontal chord 51.5 m; installed length 51.5 m

Elevation view



seabed z = -395 m (both end points at the same elevation)

Item	Value (source: specification)
OD x wall (mm)	168.3 x 14.3, X65
Inside diameter (mm)	139.7
Insulation	PU_FOAM 25 mm
Design pressure / temperature	345 bara / 120 degC
Installation	ROV-installed rigid jumper
Bends, supports, weld numbers	not defined (no as-built metrology)
Connector hubs and tolerances	not defined (vendor data)

JMP-201 J-I1: MF-201 to XT-201 (injection)

Plan view



horizontal chord 30.0 m; installed length 30.0 m

Elevation view



seabed z = -395 m (both end points at the same elevation)

Item	Value (source: specification)
OD x wall (mm)	168.3 x 14.3, X65
Inside diameter (mm)	139.7
Insulation	PU_FOAM 25 mm
Design pressure / temperature	345 bara / 80 degC
Installation	ROV-installed rigid jumper
Bends, supports, weld numbers	not defined (no as-built metrology)
Connector hubs and tolerances	not defined (vendor data)

SCREENING - screening grade: synthetic inputs, 8 open specialist hand-off(s), 5 data gap(s)

LEGEND

- production jumper
- injection jumper

GENERAL NOTES

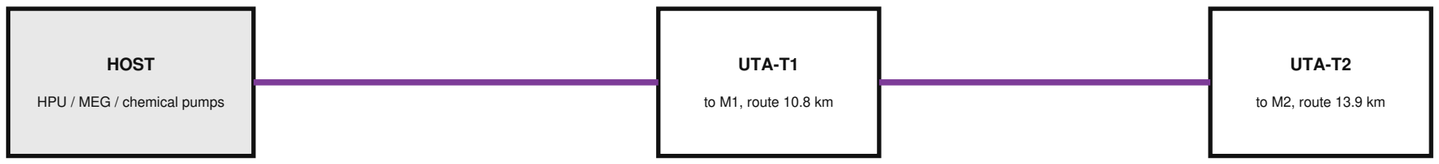
1. Conceptual end-point drawing: the jumper is a straight chord between the specified end points; bends, spans, supports and stresses are not defined.
2. No as-built metrology, weld numbering, stress results or fabrication tolerances are shown or implied.
3. Issue purpose: Issued for review (diagnostic, synthetic example); readiness SCREENING (assigned by the toolkit gates, not by a person).
4. Run reference: dataset 7a2b4b0b67d3d98726e0d58d, NeqSim workspace classes, study SURF-SYN.
5. Units: pressure bara, temperature degC, lengths m, rates kg/s. Source/model version: surf-study 1.0.

Rev	Date	Description
01	2026-10-25	First generated issue from surf_study.v1

Company: NeqSim open example		Project: Synthetic SURF demonstration	
Title: **Four-well subsea tie-back with water injection: SURF design study (synthetic example) - Jumper and tie-in details**			
Drawing number: **SURF-SYN-001-DWG-09**		Revision: 01	Status: **SCREENING**
Sheet: Sheet 9 of 15	Date: 2026-10-25	Scale: NOT TO SCALE (end-point chords; dime...	Format: A3 landscape
Drawn by: surf-study toolkit	Checked by: not assigned	Approved by: not assigned	

SURF-SYN-001-DWG-10 Umbilical architecture and conceptual cross-section

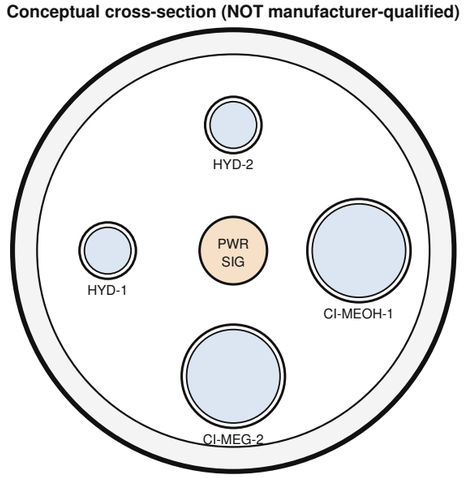
UM-101 UMB-1 - host to template umbilical terminations (functional service specification)



Service register (functional sizing; manufacturer-qualified construction is a hand-off)

Service	Kind	Consumer	Basis	Result	Assessment owner
CI-MEOH-1	chemical	M1	19 mm tube, 0.30 kg/s	330.0 bara delivered	NeqSim single-phase tube hydraulics (this study)
CI-MEG-2	chemical	M2	19 mm tube, 0.35 kg/s	168.5 bara delivered	NeqSim single-phase tube hydraulics (this study)
HYD-1	hydraulic	M1	10 mm tube	not assessed	control-system supplier (not hydraulically modelled)
HYD-2	hydraulic	M2	10 mm tube	not assessed	control-system supplier (not hydraulically modelled)
PWR-1	power	-	120 kW	not assessed	electrical specialist (voltage drop, ampacity, protection)
SIG-1	signal	-	4 channels	not assessed	control-system / telecom supplier

Conceptual cross-section (NOT manufacturer-qualified)



Tube inside diameters drawn at 0.8 mm per mm; armour, packing and cable construction are not defined.
Provenance: specification tube IDs (synthetic); layout is a conceptual arrangement.

SCREENING - screening grade: synthetic inputs, 8 open specialist hand-off(s), 5 data gap(s)

LEGEND

- umbilical (functional)
- tubes and conductors in the conceptual section

GENERAL NOTES

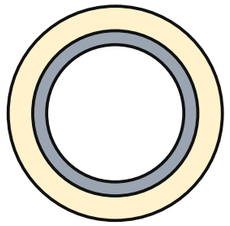
1. Chemical delivery pressure comes from a NeqSim single-phase Darcy-Weisbach solve of each tube network; hydraulic, power and signal services are registered, not modelled.
2. Electrical voltage drop, ampacity, protection, dynamic mechanics and the manufacturer cross-section require specialist tools (hand-offs HO-03 and HO-04).
3. Issue purpose: Issued for review (diagnostic, synthetic example); readiness SCREENING (assigned by the toolkit gates, not by a person).
4. Run reference: dataset 7a2b4b0b67d3d98726e0d58d, NeqSim workspace classes, study SURF-SYN.
5. Units: pressure bara, temperature degC, lengths m, rates kg/s. Source/model version: surf-study 1.0.

Rev	Date	Description
01	2026-10-25	First generated issue from surf_study.v1

Company: NeqSim open example	Project: Synthetic SURF demonstration
Title: **Four-well subsea tie-back with water injection: SURF design study (synthetic example) - Umbilical architecture and conceptual cross-section**	
Drawing number: **SURF-SYN-001-DWG-10**	Revision: 01 / Status: **SCREENING**
Sheet: Sheet 10 of 15 / Date: 2026-10-25	Scale: NOT TO SCALE (architecture); cross-sec... / Format: A3 landscape
Drawn by: surf-study toolkit	Checked by: not assigned / Approved by: not assigned

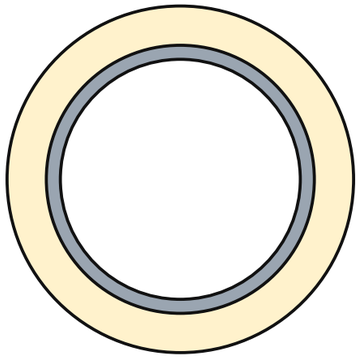
SURF-SYN-001-DWG-11 Flowline and riser thermal cross-sections

J-P1, J-P2, J-P3, J-P4, J-I1, J-I2
ID 139.7 | OD 168.3 mm | wall 14.3 mm
insulation: PU_FOAM 25 mm



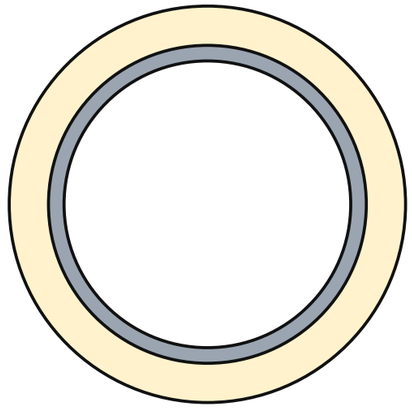
U = 1.89 W/m2K on the bore area
ambient 4 degC

FL-T2
ID 244.5 | OD 273.1 mm | wall 14.3 mm
insulation: PU_FOAM 40 mm



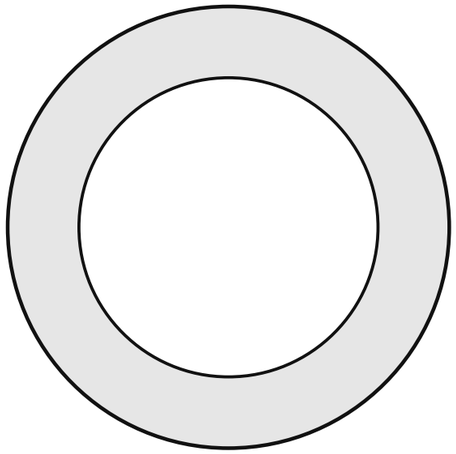
U = 1.10 W/m2K on the bore area
ambient 4 degC

FL-TRUNK
ID 292.1 | OD 323.9 mm | wall 15.9 mm
insulation: PU_FOAM 40 mm



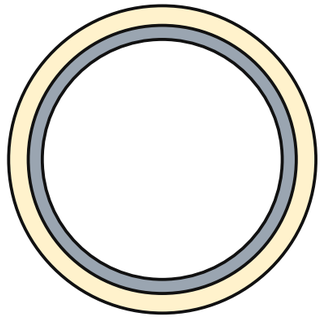
U = 1.07 W/m2K on the bore area
ambient 4 degC

RI-1
ID 304.8 | OD 450.0 mm | wall not defined (flexible)
insulation: none (explicit U-value)



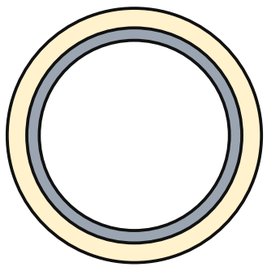
U = 2.50 W/m2K on the bore area
flexible: layers not drawn; ambient 6 degC

RI-INJ, IFL-1
ID 244.5 | OD 273.1 mm | wall 14.3 mm
insulation: PU_FOAM 20 mm



U = 2.05 W/m2K on the bore area
ambient 6 degC

IFL-T2
ID 193.7 | OD 219.1 mm | wall 12.7 mm
insulation: PU_FOAM 20 mm



U = 2.11 W/m2K on the bore area
ambient 4 degC

SCREENING - screening grade: synthetic inputs, 8 open specialist hand-off(s), 5 data gap(s)

LEGEND

☐ steel wall (grey), insulation (yellow), bore (white)

GENERAL NOTES

1. Cross-sections are drawn to the stated scale from the specification OD, wall and insulation thickness; coatings and field joints are not drawn.
2. The U-value is referenced to the inner (bore) area, the basis used by the hydraulic network; for the flexible riser it is a supplier-style value from the specification.
3. A thermal conductivity of soil is not a geotechnical resistance model and no burial effect is assumed beyond the stated ambient.
4. Issue purpose: Issued for review (diagnostic, synthetic example); readiness SCREENING (assigned by the toolkit gates, not by a person).
5. Run reference: dataset 7a2b4b0b67d3d98726e0d58d, NeqSim workspace classes, study SURF-SYN.
6. Units: pressure bara, temperature degC, lengths m, rates kg/s. Source/model version: surf-study 1.0.

Rev	Date	Description
01	2026-10-25	First generated issue from surf_study.v1

Company: NeqSim open example		Project: Synthetic SURF demonstration	
Title: **Four-well subsea tie-back with water injection: SURF design study (synthetic example) - Flowline and riser thermal cross-sections**			
Drawing number: **SURF-SYN-001-DWG-11**		Revision: 01	Status: **SCREENING**
Sheet: Sheet 11 of 15	Date: 2026-10-25	Scale: cross-sections 0.16 mm per mm (about 1:6)	Format: A3 landscape
Drawn by: surf-study toolkit	Checked by: not assigned	Approved by: not assigned	

SURF-SYN-001-DWG-12 Installation and tie-in sequence (conceptual)

S1 Install templates and manifolds

Interfaces and open analysis: lift and set; seabed preparation not assessed

Assets: T1, T2

Status: CONCEPTUAL SEQUENCE - marine-operation analyses open



S2 Install PLEM and PLET, lay trunk and infield flowlines

Interfaces and open analysis: pipelay, pull-in and tie-in methods open (HO-05)

Assets: FL-T2, FL-TRUNK, IFL-1, IFL-T2

Status: CONCEPTUAL SEQUENCE - marine-operation analyses open



S3 Install risers and umbilical

Interfaces and open analysis: riser and dynamic umbilical installation analysis open (HO-01, HO-03)

Assets: RI-1, RI-INJ

Status: CONCEPTUAL SEQUENCE - marine-operation analyses open



S4 Install jumpers (ROV tie-in)

Interfaces and open analysis: metrology and spool fabrication open (HO-08)

Assets: J-P1, J-P2, J-P3, J-P4, J-I1, J-I2

Status: CONCEPTUAL SEQUENCE - marine-operation analyses open



S5 Hydrotest, clean, gauge and dewater

Interfaces and open analysis: pre-commissioning procedures not defined

Assets: J-P1, J-P2, J-P3, J-P4, FL-T2, FL-TRUNK, RI-1, RI-INJ, IFL-1, IFL-T2, J-I1, J-I2

Status: CONCEPTUAL SEQUENCE - marine-operation analyses open



S6 Commission injection, then production

Interfaces and open analysis: start-up sequence requires transient analysis (HO-07)

Assets: J-P1, J-P2, J-P3, J-P4, FL-T2, FL-TRUNK, RI-1, RI-INJ, IFL-1, IFL-T2, J-I1, J-I2

Status: CONCEPTUAL SEQUENCE - marine-operation analyses open



SCREENING - screening grade: synthetic inputs, 8 open specialist hand-off(s), 5 data gap(s)

LEGEND

- assets installed or connected in the stage
- other assets

GENERAL NOTES

1. Conceptual sequence only: vessel capacity, lay tension and bend response, lifting, sea fastening and marine warranty acceptance are open specialist analyses (HO-05).
2. The strip plans show asset positions from the specification; no vessel, weather window or schedule is implied.
3. Pre-commissioning (testing, cleaning, drying) and commissioning steps are listed for scope only.
4. Issue purpose: Issued for review (diagnostic, synthetic example); readiness SCREENING (assigned by the toolkit gates, not by a person).
5. Run reference: dataset 7a2b4b0b67d3d98726e0d58d, NeqSim workspace classes, study SURF-SYN.
6. Units: pressure bara, temperature degC, lengths m, rates kg/s. Source/model version: surf-study 1.0.

Rev	Date	Description
01	2026-10-25	First generated issue from surf_study.v1

Company: NeqSim open example	Project: Synthetic SURF demonstration

Title: **Four-well subsea tie-back with water injection: SURF design study (synthetic example) - Installation and tie-in sequence (conceptual)**

Drawing number: **SURF-SYN-001-DWG-12**		Revision: 01	Status: **SCREENING**
Sheet: Sheet 12 of 15	Date: 2026-10-25	Scale: NOT TO SCALE (conceptual sequence; s...	Format: A3 landscape
Drawn by: surf-study toolkit	Checked by: not assigned	Approved by: not assigned	

SURF-SYN-001-DWG-13 Design-result sheet

Acceptance criteria by concept and case (value | status); governing location in the report

Criterion	A/BASE	A/TURNDOWN	A/LATE	B/BASE	B/TURNDOWN	B/LATE	C/BASE	C/TURNDOWN	C/LATE
CR-DP <= 1	0.7 PASS	0.7 PASS	0.8 PASS	0.7 PASS	0.7 PASS	0.8 PASS	FAIL	0.7 PASS	0.8 PASS
CR-HYD >= 3 dK	54.5 PASS	49.8 PASS	50.7 PASS	51.6 PASS	44.7 PASS	42.3 PASS	FAIL	47.1 PASS	50.5 PASS
CR-EROS <= 1	0.8 PASS	0.4 PASS	0.3 PASS	0.9 PASS	0.4 PASS	0.3 PASS	FAIL	0.6 PASS	0.5 PASS
CR-ARRT >= 20 C	68.7 PASS	66.7 PASS	63.9 PASS	66.5 PASS	61.6 PASS	56.9 PASS	FAIL	63.9 PASS	63.8 PASS
CR-HOST <= 60 kg/s	47.4 PASS	23.5 PASS	16.5 PASS	48.3 PASS	23.6 PASS	16.6 PASS	FAIL	22.2 PASS	15.6 PASS
CR-FRAC >= 10 bar	29.4 PASS	45.6 PASS	15.6 PASS	29.4 PASS	45.6 PASS	15.6 PASS	FAIL	45.6 PASS	15.6 PASS
CR-UMB >= 20 bar	94.1 PASS	89.9 PASS	112.5 PASS	102.7 PASS	92.1 PASS	113.3 PASS	FAIL	64.2 PASS	97.7 PASS
CR-COOL >= 8 h	30.5 PASS	32.4 PASS	36.1 PASS	33.1 PASS	32.5 PASS	36.1 PASS	FAIL	21.8 PASS	24.5 PASS

Operating envelope over all cases, reference concept A (bara, degC)

Line	Tag	p max	p min	T max	T min	design p/T
J-P1	JMP-101	77.5	55.3	79.9	79.9	345 / 120
J-P2	JMP-102	77.5	55.3	79.9	79.9	345 / 120
J-P3	JMP-103	78.7	56.0	79.9	79.9	345 / 120
J-P4	JMP-104	78.6	56.0	79.9	79.9	345 / 120
FL-T2	FL-102	78.5	55.3	79.5	74.3	345 / 120
FL-TRUNK	FL-101	77.3	47.0	78.1	66.9	345 / 120
RI-1	RI-101	71.3	36.0	70.6	64.2	345 / 120
RI-INJ	RI-201	266.1	200.0	30.0	29.1	345 / 80
IFL-1	FL-201	266.1	233.9	30.0	28.8	345 / 80
IFL-T2	FL-202	264.5	233.9	30.0	29.3	345 / 80
J-I1	JMP-201	263.9	233.8	30.0	30.0	345 / 80
J-I2	JMP-202	264.5	234.4	30.0	30.0	345 / 80

Host arrival conditions and injection pump power (kg/s per phase)

Case	p bara	T degC	m kg/s	gas	oil	water	pump kW
BASE	40.0	68.7	47.4	29.8	16.3	1.3	2835
TURNDOWN	60.0	66.7	23.5	14.2	8.5	0.7	2098
LATE	35.0	63.9	16.5	7.3	7.5	1.7	2430

SCREENING - screening grade: synthetic inputs, 8 open specialist hand-off(s), 5 data gap(s)

LEGEND

GENERAL NOTES

1. Values are the solved steady NeqSim results of each concept and case; PASS/FAIL follows the specification criteria and a non-converged case is never PASS.
2. A criterion that could not be assessed is shown as NOT ASSESSED and blocks ranking of the concept.
3. Issue purpose: Issued for review (diagnostic, synthetic example); readiness SCREENING (assigned by the toolkit gates, not by a person).
4. Run reference: dataset 7a2b4b0b67d3d98726e0d58d, NeqSim workspace classes, study SURF-SYN.
5. Units: pressure bara, temperature degC, lengths m, rates kg/s. Source/model version: surf-study 1.0.

Rev	Date	Description
01	2026-10-25	First generated issue from surf_study.v1

Company: NeqSim open example	Project: Synthetic SURF demonstration
Title: **Four-well subsea tie-back with water injection: SURF design study (synthetic example) - Design-result sheet**	
Drawing number: **SURF-SYN-001-DWG-13**	Revision: 01 / Status: **SCREENING**
Sheet: Sheet 13 of 15 / Date: 2026-10-25	Scale: see sheet notes / Format: A3 landscape
Drawn by: surf-study toolkit / Checked by: not assigned	Approved by: not assigned

SURF-SYN-001-DWG-14 Concept comparison

Concept A - daisy chain

Template 2 flows to template 1 through a short infield flowline; one shared trunk line carries all four wells to the host.



Feasibility: FEASIBLE (SELECTED)

SURF CAPEX: 586 MUSD (USD, 2025, class CLASS_4 screening)

Production line length: 13.8 km

Failed criteria: none

Open criteria: none

Concept B - dual direct flowlines

Template 2 has its own direct flowline to the PLEM; template 1 uses the trunk. Two long lines share the riser base.



Feasibility: FEASIBLE

SURF CAPEX: 605 MUSD (USD, 2025, class CLASS_4 screening)

Production line length: 23.8 km

Failed criteria: none

Open criteria: none

Concept C - daisy chain with an 8-inch trunk

As concept A but the trunk is 219.1 x 12.7 mm instead of 323.9 x 15.9 mm: a cheaper line that is checked for feasibility before it can be ranked.



Feasibility: INFEASIBLE

SURF CAPEX: 579 MUSD (USD, 2025, class CLASS_4 screening)

Production line length: 13.8 km

Failed criteria: CR-DP/BASE, CR-HYD/BASE, CR-EROS/BASE

Open criteria: none

SCREENING - screening grade: synthetic inputs, 8 open specialist hand-off(s), 5 data gap(s)

LEGEND

GENERAL NOTES

1. Feasibility is assessed before cost: a concept is ranked on cost only if every acceptance criterion passes in every case.
2. CAPEX is the existing SURFCostEstimator screening estimate; wells and host modifications are outside the scope and cost is not an estimate-class claim.
3. Issue purpose: Issued for review (diagnostic, synthetic example); readiness SCREENING (assigned by the toolkit gates, not by a person).
4. Run reference: dataset 7a2b4b0b67d3d98726e0d58d, NeqSim workspace classes, study SURF-SYN.
5. Units: pressure bara, temperature degC, lengths m, rates kg/s. Source/model version: surf-study 1.0.

Rev	Date	Description
01	2026-10-25	First generated issue from surf_study.v1

Company: NeqSim open example	Project: Synthetic SURF demonstration
Title: **Four-well subsea tie-back with water injection: SURF design study (synthetic example) - Concept comparison**	
Drawing number: **SURF-SYN-001-DWG-14**	Revision: 01 / Status: **SCREENING**
Sheet: Sheet 14 of 15 / Date: 2026-10-25 / Scale: strip plans NOT TO SCALE in the transve... / Format: A3 landscape	
Drawn by: surf-study toolkit / Checked by: not assigned / Approved by: not assigned	

SURF-SYN-001-DWG-15 Quantities, interfaces and drawing register

Line register (reference concept A)

Line	Tag	Kind	Service	Route m	Installed m	OD x WT mm	ID mm	Insulation
J-P1	JMP-101	jumper	production	51	51	168.3 x 14.3	140	PU_FOAM 25 mm
J-P2	JMP-102	jumper	production	51	51	168.3 x 14.3	140	PU_FOAM 25 mm
J-P3	JMP-103	jumper	production	51	51	168.3 x 14.3	140	PU_FOAM 25 mm
J-P4	JMP-104	jumper	production	51	51	168.3 x 14.3	140	PU_FOAM 25 mm
FL-T2	FL-102	flowline	production	3040	3040	273.1 x 14.3	245	PU_FOAM 40 mm
FL-TRUNK	FL-101	trunkline	production	10002	10002	323.9 x 15.9	292	PU_FOAM 40 mm
RI-1	RI-101	riser	production	599	599	ID 305	305	none (explicit U-value)
RI-INJ	RI-201	riser	injection	636	636	273.1 x 14.3	245	PU_FOAM 20 mm
IFL-1	FL-201	flowline	injection	9968	9968	273.1 x 14.3	245	PU_FOAM 20 mm
IFL-T2	FL-202	flowline	injection	3040	3040	219.1 x 12.7	194	PU_FOAM 20 mm
J-I1	JMP-201	jumper	injection	30	30	168.3 x 14.3	140	PU_FOAM 25 mm
J-I2	JMP-202	jumper	injection	30	30	168.3 x 14.3	140	PU_FOAM 25 mm

Drawing register (this package)

Sheet	Title
DWG-01	Overall field layout plan
DWG-02	SURF process and network schematic
DWG-03	Production process-flow and functional sheet
DWG-04	Injection process-flow and functional sheet
DWG-05	Production route longitudinal profile
DWG-06	Injection route longitudinal profile
DWG-07	Riser configuration and elevation (concept schematic)
DWG-08	Template, manifold, PLEM and PLET arrangement
DWG-09	Jumper and tie-in details
DWG-10	Umbilical architecture and conceptual cross-section
DWG-11	Flowline and riser thermal cross-sections
DWG-12	Installation and tie-in sequence (conceptual)
DWG-13	Design-result sheet
DWG-14	Concept comparison
DWG-15	Quantities, interfaces and drawing register

Equipment counts (specification = design basis = cost basis)

Equipment kind	Count
host	1
injection manifold	2
injection source	1
injection well	2
junction	1
manifold	2
plem	1
plet	1
production well	4
templates	2

Material take-off summary (geometry-derived)

MTO item	Quantity
steel pipe (derived)	2664 t
insulation (derived)	822 m3
umbilical	13.9 km

Reconciliation of model, register, design basis and cost: 19 checks, 19 PASS.

SCREENING - screening grade: synthetic inputs, 8 open specialist hand-off(s), 5 data gap(s)

LEGEND

GENERAL NOTES

1. Every quantity is derived from the canonical specification geometry; installed length = route length x (1 + stated allowance), used once.
2. The cost basis and the line register agree on counts and lengths (see the validation report); the estimate itself is a Class 4 screening value from the existing estimator.
3. Issue purpose: Issued for review (diagnostic, synthetic example); readiness SCREENING (assigned by the toolkit gates, not by a person).
4. Run reference: dataset 7a2b4b0b67d3d98726e0d58d, NeqSim workspace classes, study SURF-SYN.
5. Units: pressure bara, temperature degC, lengths m, rates kg/s. Source/model version: surf-study 1.0.

Rev	Date	Description
01	2026-10-25	First generated issue from surf_study.v1

Company: NeqSim open example	Project: Synthetic SURF demonstration
Title: **Four-well subsea tie-back with water injection: SURF design study (synthetic example) - Quantities, interfaces and drawing register**	
Drawing number: **SURF-SYN-001-DWG-15**	Revision: 01 / Status: **SCREENING**
Sheet: Sheet 15 of 15 / Date: 2026-10-25	Scale: see sheet notes / Format: A3 landscape
Drawn by: surf-study toolkit / Checked by: not assigned	Approved by: not assigned